

Multiple Choice

1. **Answer:** B

Options: A) $-\frac{v^2}{k}$; B) $-\frac{3}{4} \frac{v^2}{k}$; C) $-\frac{7}{4} \frac{v^2}{k}$; D) $\frac{1}{4} \frac{v^2}{k}$; E) $-\frac{1}{4} \frac{v^2}{k}$

Note: Correct option: B - $-\frac{3}{4} \frac{v^2}{k}$

2. **Answer:** A

Options: A) 2.50; B) 0.89; C) 2.23; D) 2.75; E) 4.5

Note: Correct option: A - 2.50

3. **Answer:** A

Options: A) $\frac{2}{\sqrt{2}}R$; B) $\frac{3}{\sqrt{2}}R$; C) R ; D) $\sqrt{2}R$; E) $\frac{1}{\sqrt{3}}R$

Note: Correct option: A - $\frac{2}{\sqrt{2}}R$

4. **Answer:** B

Options: A) $97\mu\text{C}$; B) $37\mu\text{C}$; C) $77\mu\text{C}$; D) $87\mu\text{C}$; E) $17\mu\text{C}$

Note: Correct option: B - $37\mu\text{C}$

5. **Answer:** C

Options: A) 40 km/h; B) 60 km/h; C) 50 km/h; D) 10 km/h; E) 90 km/h

Note: Correct option: C - 50 km/h

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'decreases'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '350 cm'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'defying gravity through an internal mechanism.'

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 0.045. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** 75 joules. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.